SWARNA BHARATHI
INSTITUTE OF SCIENCE & TECHNOLOGY (SBIT)
UGC -Autonomous
(Approved by AICTE, Affiliated to JNTUH, Accredited by NAAC with A+)
H No. 5-7-200/12, Pakabanda Street, Khammam,TS- 507002

Presentation on

# RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation

**Presented By :**

1. G. LOHITHA - 23M61A0521
2. CH. NEELIMA - 23M61A0516
3. P. KARUNA SRI - 23M61A0545
4. R. SOWJANYA - 23M61A0550
5. N. DHARANI - 23M61A0542

**Guide**

**Mrs. Dr N Savitha**

**Batch No : 14** **Date:**

**Dept. of Computer Science & Engineering( Data Science)**

SWARNA BHARATHI
INSTITUTE OF SCIENCE & TECHNOLOGY (SBIT)
UGC -Autonomous
(Approved by AICTE, Affiliated to JNTUH, Accredited by NAAC with A+)
H No. 5-7-200/12, Pakabanda Street, Khammam,TS- 507002

# ABSTRACT

## AI-Based Polysemy Disambiguation & Automatic Title Generation

### 1. PROBLEM STATEMENT

Polysemous words can have multiple meanings, making it difficult for NLP systems to correctly understand text context and generate suitable titles.

### 2. OBJECTIVE

To improve contextual understanding by resolving polysemous words and automatically generating meaningful, context-appropriate titles from textual content.

### 3. PROPOSED SOLUTION

A RAG-enhanced Transformer framework is proposed to retrieve relevant external knowledge, disambiguate polysemous words, and generate suitable titles based on the given content.

### 4. EXPECTED OUTCOME

The system is expected to provide better contextual understanding, accurate polysemy disambiguation, and relevant, meaningful titles for diverse textual content.

### METHODOLOGY




KEYWORDS | RAG • Transformer • NLP • Polysemy • Contextual Understanding • Disambiguation

Dept. of Computer Science & Engineering

SWARNA BHARATHI
INSTITUTE OF SCIENCE & TECHNOLOGY (SBIT)
UGC -Autonomous
(Approved by AICTE, Affiliated to JNTUH, Accredited by NAAC with A+)
H No. 5-7-200/12, Pakabanda Street, Khammam,TS- 507002

# SYSTEM ANALYSIS

## 1. Identification of Need

- Short headlines lack sufficient context.
- Polysemous words cause semantic ambiguity.
- Existing models may misclassify word meanings.

## 2. Technical Feasibility

- RoBERTa and XLM-RoBERTa for contextual analysis.
- Knowledge distillation for lightweight models.
- Soft-voting ensemble for robust predictions.

## 3. Economic Feasibility

- Open-source tools and libraries.
- Reduced computing and deployment costs.
- Compatibility with standard hardware.

## 4. Operational Feasibility

- Flask REST API integration.
- Independent knowledge-base updates.
- Modular model deployment and maintenance.

## 5. User Requirements

- Fast and accurate word-sense identification.
- Reliable semantic classification.
- Independent model and knowledge-base management.

## 6. Scope of the System

**In-Scope**

- ✓ Headline preprocessing
- ✓ Sense retrieval
- ✓ Contextual disambiguation
- ✓ Ensemble classification
- ✓ API integration

**Out-of-Scope**

- ✗ Full-document translation
- ✗ Long-form summarization
- ✗ Image/video analysis
- ✗ Live web crawling during inference

## EXISTING SYSTEM

- **RAG-DistilBERT:** Uses DistilBERT for polysemy disambiguation in digital news headlines.
- **Knowledge Base:** WordNet provides word senses, definitions, POS tags, and examples.
- **Sense Retrieval:** Sentence Transformers generate embeddings, which are indexed using FAISS.
- **Hybrid Scoring:** Lesk and Jaccard similarity help identify relevant word senses.
- **Classification:** The augmented headline is processed by DistilBERT to predict the word sense.
- **Workflow:**
  Input Headline → WordNet → FAISS Retrieval → Hybrid Scoring → DistilBERT Classification → Predicted Word Sense

## DISADVANTAGES

- High Training Latency
- Limited Knowledge Coverage
- Context Sparsity
- Training Instability
- Misclassification

# PROPOSED SYSTEM

| Component / Stage | Description |
|---|---|
| **Input & Preprocessing** | Cleans and tokenizes headlines using RoBERTa's Byte-Pair Encoding (BPE). |
| **Knowledge Base & Retriever** | Uses an external knowledge base (knowledge_base.csv) and a TF-IDF retriever for efficient context retrieval. |
| **Teacher Models (RAG)** | Uses RAG-RoBERTa and RAG-XLM-RoBERTa for contextual and cross-lingual understanding. |
| **Knowledge Distillation** | Transfers knowledge from the teacher model to a lightweight DistilRoBERTa student model using Cross-Entropy and KL Divergence losses. |
| **Ensemble Soft-Voting** | Combines prediction probabilities from RAG-RoBERTa, RAG-XLM-RoBERTa, and RAG-DistilRoBERTa-KD. |

# ADVANTAGES

- Reduced Training Latency
- Reduced Misclassification
- Improved Contextual Understanding
- Scalability and Maintainability
- Cross-Lingual Robustness
- Dynamic Knowledge Enhancement

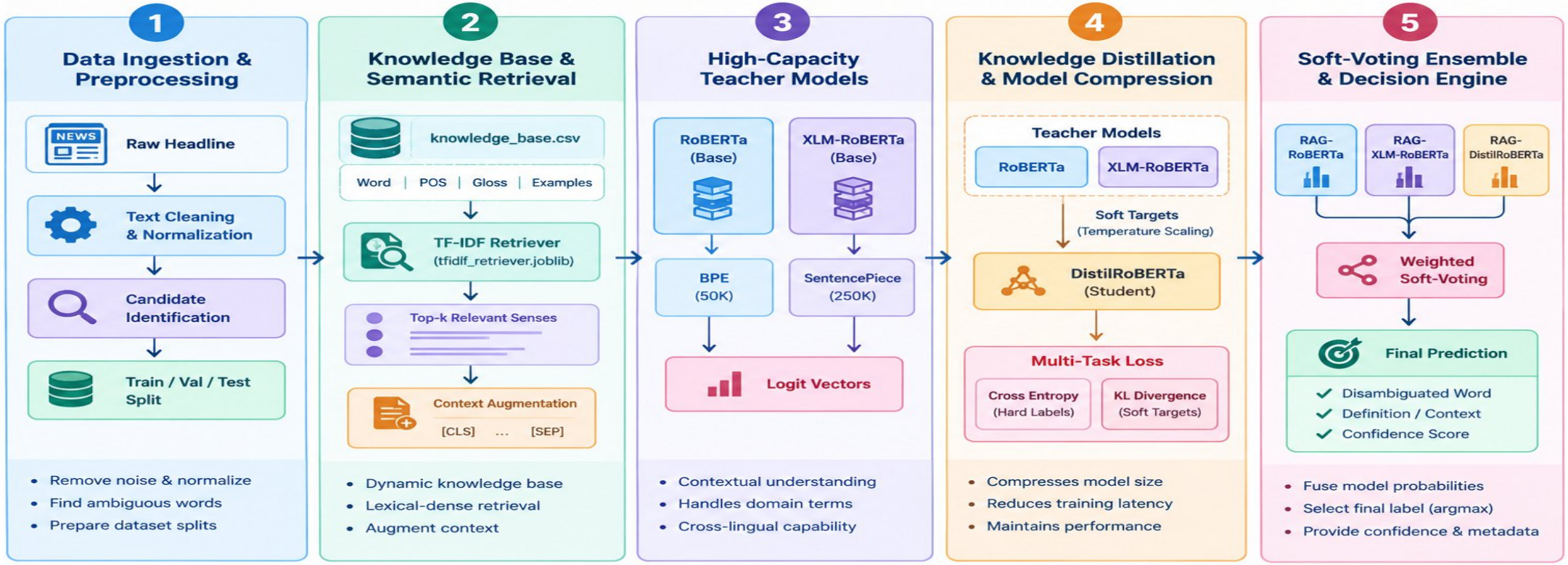
# SWARNA BHARATHI
## INSTITUTE OF SCIENCE & TECHNOLOGY (SBIT)
**UGC -Autonomous**

*(Approved by AICTE, Affiliated to JNTUH, Accredited by NAAC with A+)*

H No. 5-7-200/12, Pakabanda Street, Khammam,TS- 507002


# MODULES EXPLANATION

Five modular subsystems for efficient, accurate and scalable polysemy disambiguation




**Dept. of Computer Science & Engineering**